

1. OBJECTIVE

To write and execute a C program using switch-case and loops to perform operations like temperature conversion, prime number checking, and factorial calculation.

2. ALGORITHM

1. Start the program.
2. Display the options menu (1. Temp F to C, 2. Temp C to F, 3. Prime Check, 4. Factorial, 5. Exit).
3. Take choice input from user.
4. Execute corresponding block using switch-case.
5. Repeat using do-while loop until choice is 5.
6. Stop.

3. SOURCE CODE

```
#include <stdio.h>

void fahrenheitToCelsius() {
    float f, c;
    printf("
Enter temperature in Fahrenheit: ");
    scanf("%f", &f);
    c = (f - 32.0) * 5.0 / 9.0;
    printf("Result: %.2f F = %.2f C
", f, c);
}

void celsiusToFahrenheit() {
    float c, f;
    printf("
Enter temperature in Celsius: ");
    scanf("%f", &c);
    f = (c * 9.0 / 5.0) + 32.0;
    printf("Result: %.2f C = %.2f F
", c, f);
}

void checkPrime() {
    int num, isPrime = 1;
    printf("
Enter a positive integer: ");
    scanf("%d", &num);

    if (num <= 1) {
        isPrime = 0;
    } else {
        for (int i = 2; i * i <= num; i++) {
            if (num % i == 0) {
                isPrime = 0;
                break;
            }
        }
    }

    if (isPrime)
        printf("Result: %d is a Prime Number.
", num);
}
```

```

        else
            printf("Result: %d is NOT a Prime Number.
", num);
    }

void calculateFactorial() {
    int num;
    unsigned long long fact = 1;
    printf("
Enter a non-negative integer: ");
    scanf("%d", &num);

    if (num < 0) {
        printf("Error: Factorial of negative numbers is undefined.
");
        return;
    }

    for (int i = 1; i <= num; i++) {
        fact *= i;
    }
    printf("Result: Factorial of %d = %llu
", num, fact);
}

int main() {
    int choice;
    do {
        printf("
*** MENU ***
");
        printf("1. Fahrenheit to Celsius
");
        printf("2. Celsius to Fahrenheit
");
        printf("3. Check Prime Number
");
        printf("4. Calculate Factorial
");
        printf("5. Exit
");
        printf("Enter choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                fahrenheitToCelsius();
                break;
            case 2:
                celsiusToFahrenheit();
                break;
            case 3:
                checkPrime();
                break;
            case 4:
                calculateFactorial();
                break;
            case 5:
                printf("
Exiting program...
");
                break;
            default:

```

```

        printf("
Invalid choice! Try again.
");
    }
} while (choice != 5);

return 0;
}

```

4. TEST CASES

S.No	Test Input	Expected Output	Result
1	Choice 1, Temp = 98.6	37.00 C	Pass
2	Choice 2, Temp = 0	32.00 F	Pass
3	Choice 3, Num = 29	Prime Number	Pass
4	Choice 4, Num = 5	Factorial = 120	Pass
5	Choice 5	Exit Program	Pass

5. OUTPUT SCREENSHOT

```
$ gcc program.c -o program && ./program
```

```
*** MENU ***
```

1. Fahrenheit to Celsius
2. Celsius to Fahrenheit
3. Check Prime Number
4. Calculate Factorial
5. Exit

```
Enter choice: 1
```

```
Enter temperature in Fahrenheit: 98.6
```

```
Result: 98.60 F = 37.00 C
```

```
*** MENU ***
```

1. Fahrenheit to Celsius
2. Celsius to Fahrenheit
3. Check Prime Number
4. Calculate Factorial
5. Exit

```
Enter choice: 3
```

```
Enter a positive integer: 29
```

```
Result: 29 is a Prime Number.
```

```
*** MENU ***
```

1. Fahrenheit to Celsius
2. Celsius to Fahrenheit
3. Check Prime Number
4. Calculate Factorial
5. Exit

```
Enter choice: 5
```

```
Exiting program...
```

```
$
```

6. CONCLUSION

The program was written, compiled using GCC, and executed successfully. All options were verified.